LAB EXPERIMENT RECORD

SET 15: Product Sales Analysis

Department	AIDS
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R
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Aim

To analyze and visualize monthly product sales data using grouped bar charts, stacked area charts, tables, and an interactive dashboard in RStudio.

Dataset

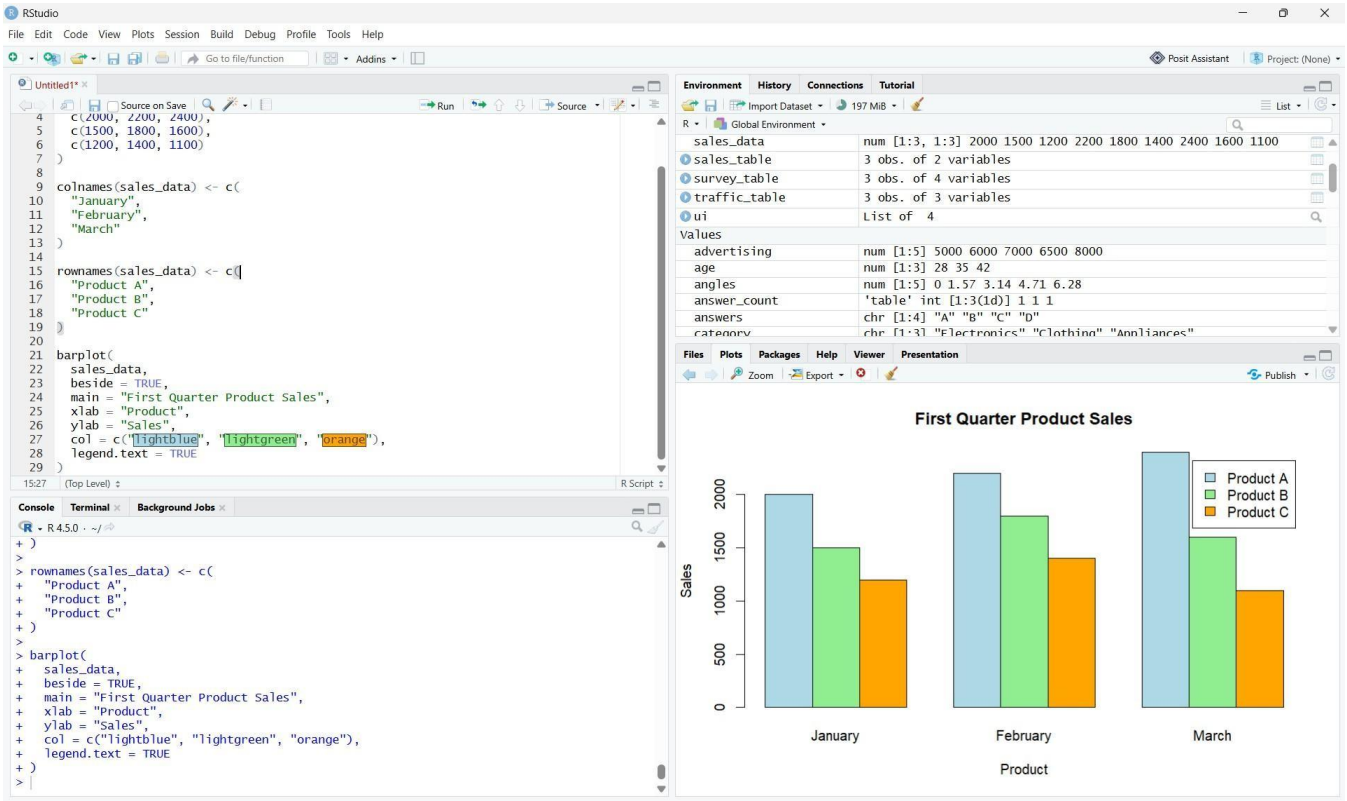
Product ID	Product Name	January	February	March
1	Product A	2000	2200	2400
2	Product B	1500	1800	1600
3	Product C	1200	1400	1100

PROGRAM 1: Grouped Bar Chart

To compare the first-quarter monthly sales of each product using a grouped bar chart

```
# # Set 15 Q1 - Grouped Bar Chart
sales_data <- rbind(
  c(2000, 2200, 2400),
  c(1500, 1800, 1600),
  c(1200, 1400, 1100)
)
colnames(sales_data) <- c(
  "January",
  "February",
  "March"
)
rownames(sales_data) <- c(
  "Product A",
  "Product B",
  "Product C"
)
```

```
barplot(
  sales_data,
  beside = TRUE,
  main = "First Quarter Product Sales",
  xlab = "Product",
  ylab = "Sales",
  col = c("lightblue", "lightgreen", "orange"),
  legend.text = TRUE
) col = c("lightblue", "lightgreen", "orange")
)
```



Observation:

- **Product A has the highest sales in all three months.**
- **Product B has medium sales.**
- **Product C has the lowest sales.**

PROGRAM 2: Stacked Area Chart

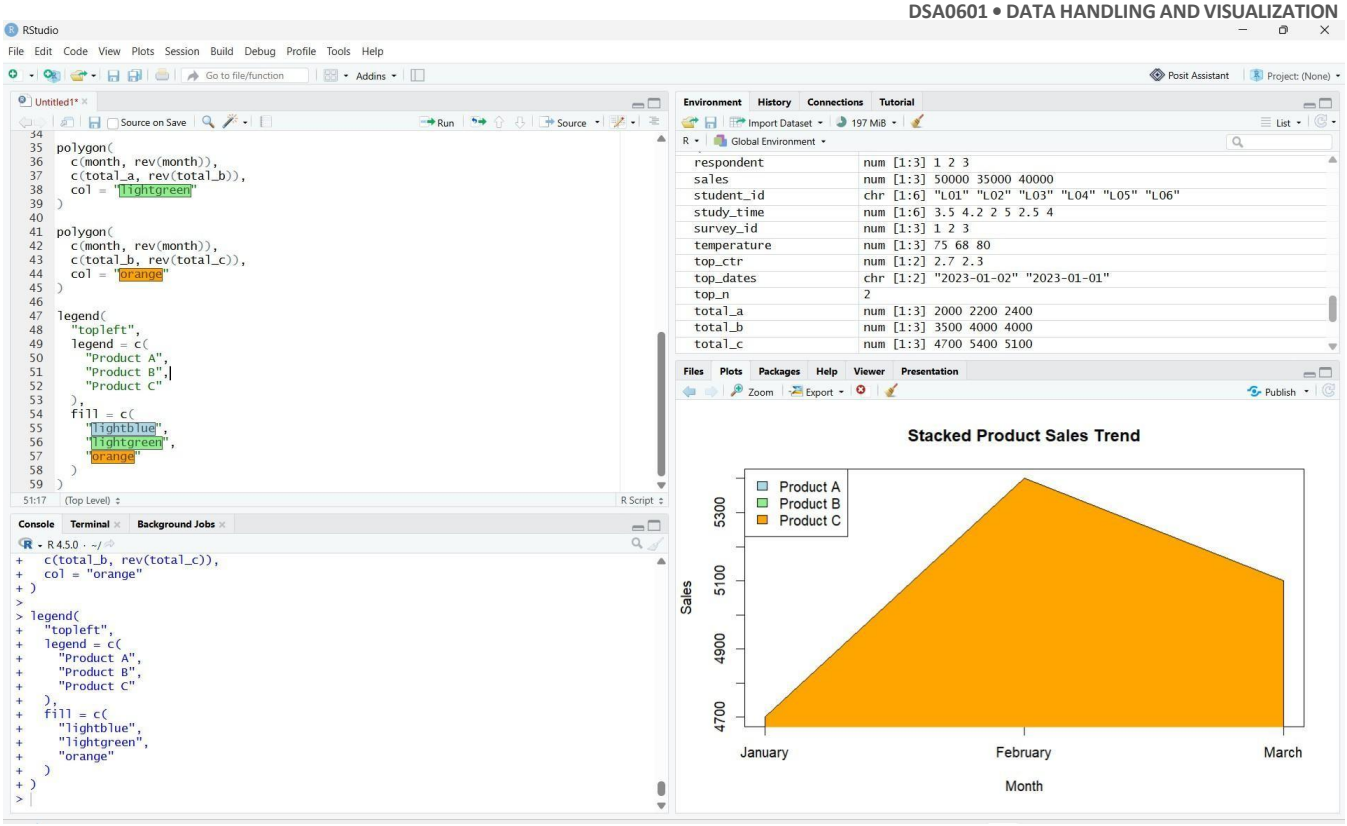
To represent the overall sales trend of all products over January, February and March

```
# # Set 15 Q2 - Stacked Area Chart


month <- 1:3


product_a <- c(2000, 2200, 2400)
product_b <- c(1500, 1800, 1600)
product_c <- c(1200, 1400, 1100)
total_a <- product_a
total_b <- product_a + product_b
total_c <- product_a + product_b + product_c
plot(
  month,
  total_c,
  type = "n",
  xaxt = "n",
  main = "Stacked Product Sales Trend",
  xlab = "Month",
  ylab = "Sales"
)
axis(
  1,
  at = 1:3,
  labels = c("January", "February", "March")
)
polygon(
  c(month, rev(month)),
  c(rep(0, 3), rev(total_a)),
  col = "lightblue"
)
polygon(
  c(month, rev(month)),
  c(total_a, rev(total_b)),
```

```
  col = "lightgreen"
)
polygon(
  c(month, rev(month)),
  c(total_b, rev(total_c)),
  col = "orange"
)
legend(
  "topleft",
  legend = c(
    "Product A",
    "Product B",
    "Product C"
  ),
  fill = c(
    "lightblue",
    "lightgreen",
    "orange"
  )
)
```



Observation:

TThe total sales are:

- **January = 4700**
- **February = 5400**
- **March = 5100**

Overall sales increase from January to February and slightly decrease in March.

PROGRAM 3: Monthly Sales Table

To display the monthly sales data for each product in tabular form

```
# Set 15 Q3 - Monthly Sales Table

product_id <- c(1, 2, 3)

product_name <- c(

  "Product A",

  "Product B",

  "Product C"

)

january <- c(2000, 1500, 1200)
```

```
february <- c(2200, 1800, 1400)

march <- c(2400, 1600, 1100)

sales_table <- data.frame(

  Product_ID = product_id,

  Product_Name = product_name,

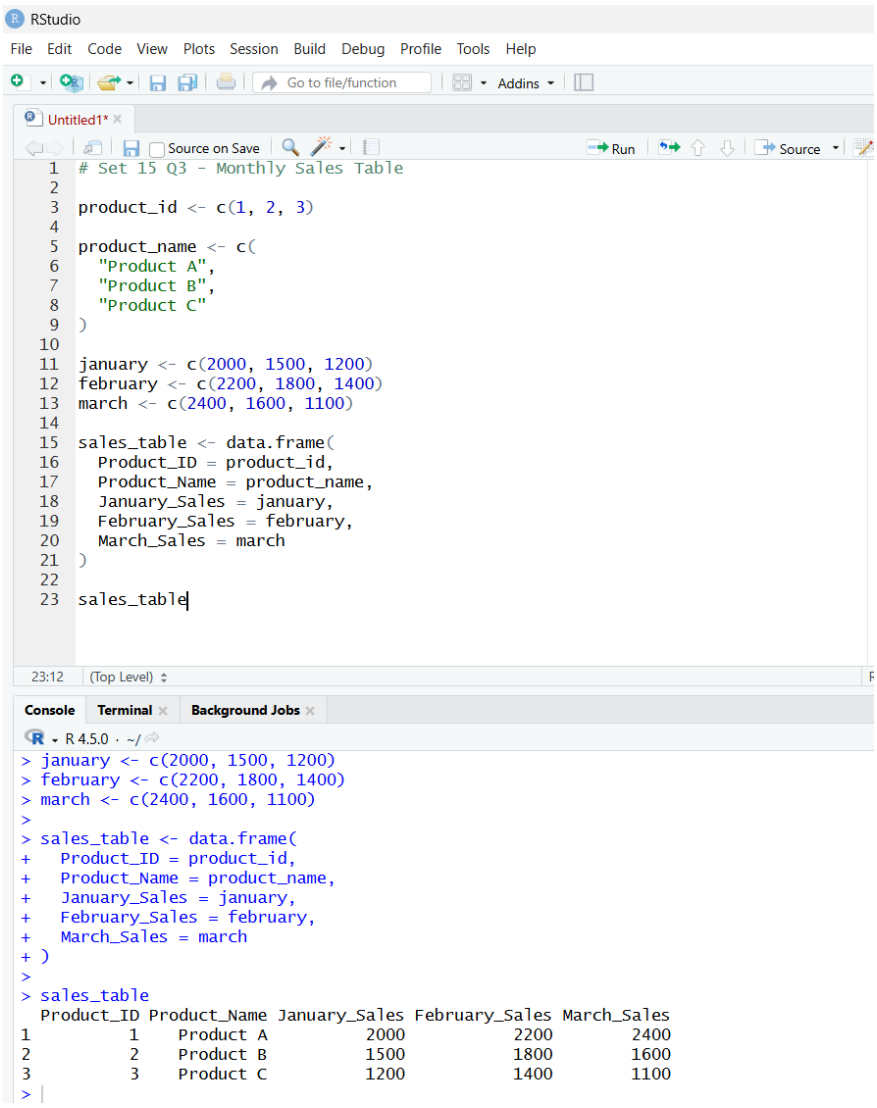
  January_Sales = january,

  February_Sales = february,

  March_Sales = march

)

sales_table
```



Observation

The monthly sales data of all products was successfully displayed in a table.

PROGRAM 4: Product Sales Dashboard

```
    library(shiny)

product <- c(
  "Product A",
  "Product B",
  "Product C"
)

january <- c(2000, 1500, 1200)
february <- c(2200, 1800, 1400)
march <- c(2400, 1600, 1100)

ui <- fluidPage(

  titlePanel("Monthly Product Sales Dashboard"),

  tabsetPanel(

    tabPanel(
      "Grouped Bar Chart",
      plotOutput("bar")
    ),

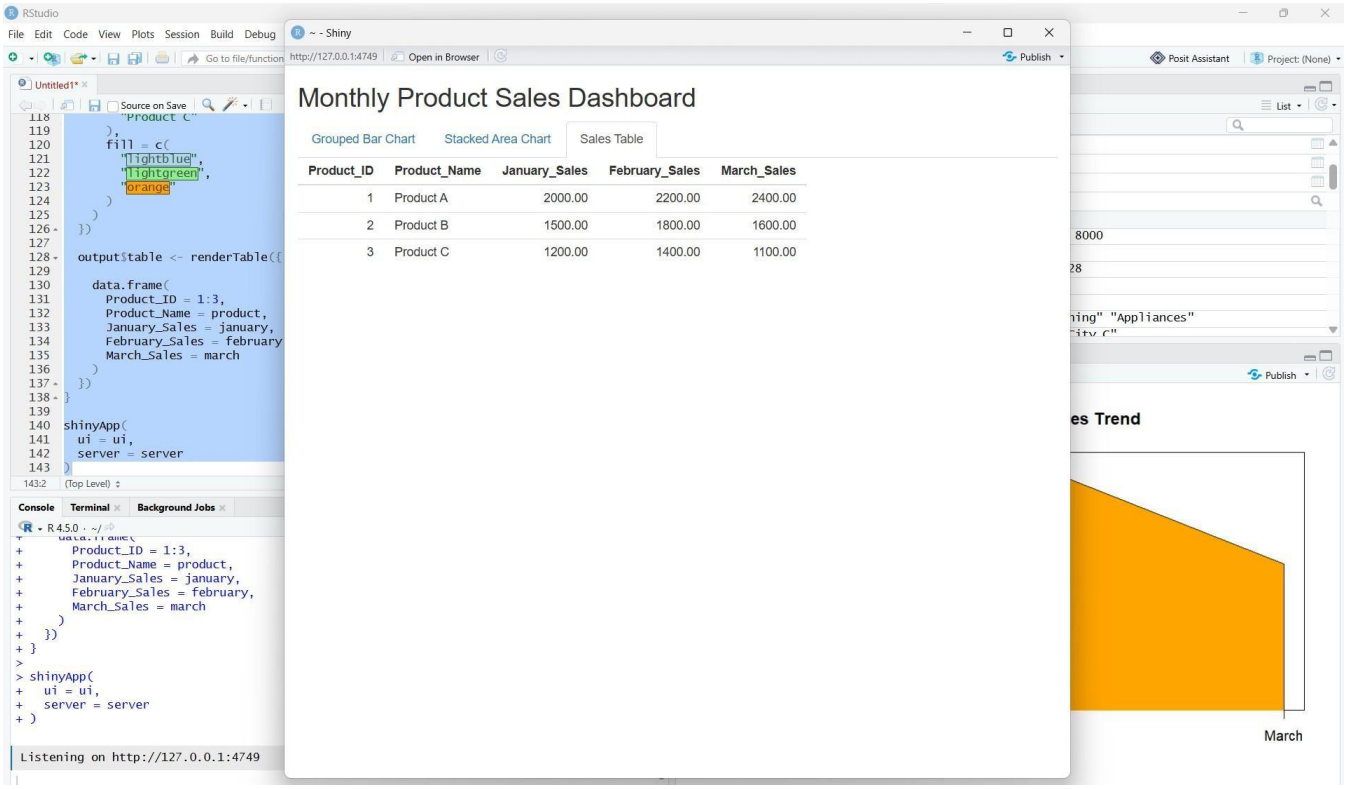



  output$table <- renderTable({

    data.frame(
      Product_ID = 1:3,
      Product_Name = product,
      January_Sales = january,
      February_Sales = february,
      March_Sales = march
    )
  })
```

```
}


shinyApp(
  ui = ui,
  server = server
)shinyApp(ui = ui, server = server)shinyApp(ui = ui, server = server))
```



Output 4: Interactive Monthly Sales Dashboard

Observation: An interactive product sales dashboard containing the grouped bar chart, stacked area chart and monthly sales table was successfully developed using RStudio Shiny..

Procedure

1. **Create vectors for Product ID, Product Name, January Sales, February Sales, and March Sales.**
2. **Use a grouped bar chart to compare the sales of each product during the first quarter.**
3. **Create a stacked area chart using the monthly sales values to visualize the overall sales trend of all products.**
4. **Create a data frame/table to display the monthly sales information for each product.**
5. **Develop an interactive Shiny dashboard in RStudio combining the grouped bar chart, stacked area chart, and sales table.**
6. **Run the programs in RStudio and observe the visualizations and sales trends.**

Result:

The monthly product sales were successfully analyzed using different visualizations and an interactive dashboard, making it easy to compare products and understand their sales trends.